| | |
|---|---|
| $a \coloneqq E$ | **Assignment**: compute E and write it to a |

$CONWHILE$ sequential commands:

$C \in Com ::= a \coloneqq E$ [assignment]
$\mid a \coloneqq x$ [(memory) read]
$\mid x \coloneqq a$ [(memory) write]
$\mid a \coloneqq CAS(x, E, E) \mid FAA(x, E)$ [(memory) RMWs]
$\mid skip \mid C; C \mid while\ B\ do\ C$
$\mid if\ B\ then\ C\ else\ C$

$CONWHILE$ concurrent programs: $P \in Prog \triangleq Tid \rightarrow Com$

For n-threaded program P: $C_1 || C_2 || \dots || C_n$ with $dom(P) = \{\tau_1, \dots, \tau_n\}$ and $P(\tau_i) = C_i$ for $i \in \{1, \dots, n\}$.

**SC Configurations:**

- **Shared memory:** $M \in Mem \triangleq Loc \rightarrow Val$. $Val$: set of all values, including integer and Boolean values.
- **Store:** $s \in Store \triangleq Reg \rightarrow Val$.
- **Store Map:** $S \in SMap \triangleq Tid \rightarrow Store$. $s = S(\tau)$: the store map of $\tau$.

$s(b)$: the value of register $b$ for thread $\tau$.

- **SC configuration:** $(P, S, M)$